

Joint Knowledge Distillation and Tensor-Based Compression of Deep Neural Networks

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1 Introduction

Deep neural networks — ResNet, WideResNet, VGG — deliver high accuracy but carry millions of parameters and billions of FLOPs per inference, far exceeding the memory and power budgets of edge platforms such as autonomous drones, IoT sensors and medical devices. This bottleneck is the *memory wall* [1]: fetching model parameters from DRAM eclipses available compute, making aggressive compression an engineering imperative for practical edge AI.

Two compression families exist yet each fails in isolation. **Tensor decomposition** (CP [2], Tucker [3]) reduces structural size but causes vanishing gradients and accuracy collapse in the sequential *decompose-then-fine-tune* pipeline. **Knowledge distillation** [5] guides a compact student with a teacher's soft outputs but collapses at high compression ratios due to the *capacity gap* [6] and becomes structurally ill-defined once the student's layers are tensor-factorized.

2 Problem Definition

Given a pre-trained teacher $\mathcal{T}$ and labelled dataset $\mathcal{D}$, jointly construct and train two compressed students S_{CP} , S_{Tuck} from scratch to minimise the combined distillation loss:

$$\mathcal{L} = \sum_{i \in \{CE, KL, AT\}} (e^{-s_i} \mathcal{L}_i + s_i)$$

subject to:

- Per-layer ranks $\{r_i\}$ derived by **EVBMF** on $\mathcal{T}$'s weight spectrum — automatic, layer-adaptive, no manual tuning.
- Hint points $\mathcal{H} = \{h_1, \dots, h_K\}$ selected by **PURSUhInT** clustering of $\mathcal{T}$'s layer representations.
- Log-variance weights s_i learned online (Kendall uncertainty) — zero hand-tuned loss coefficients.

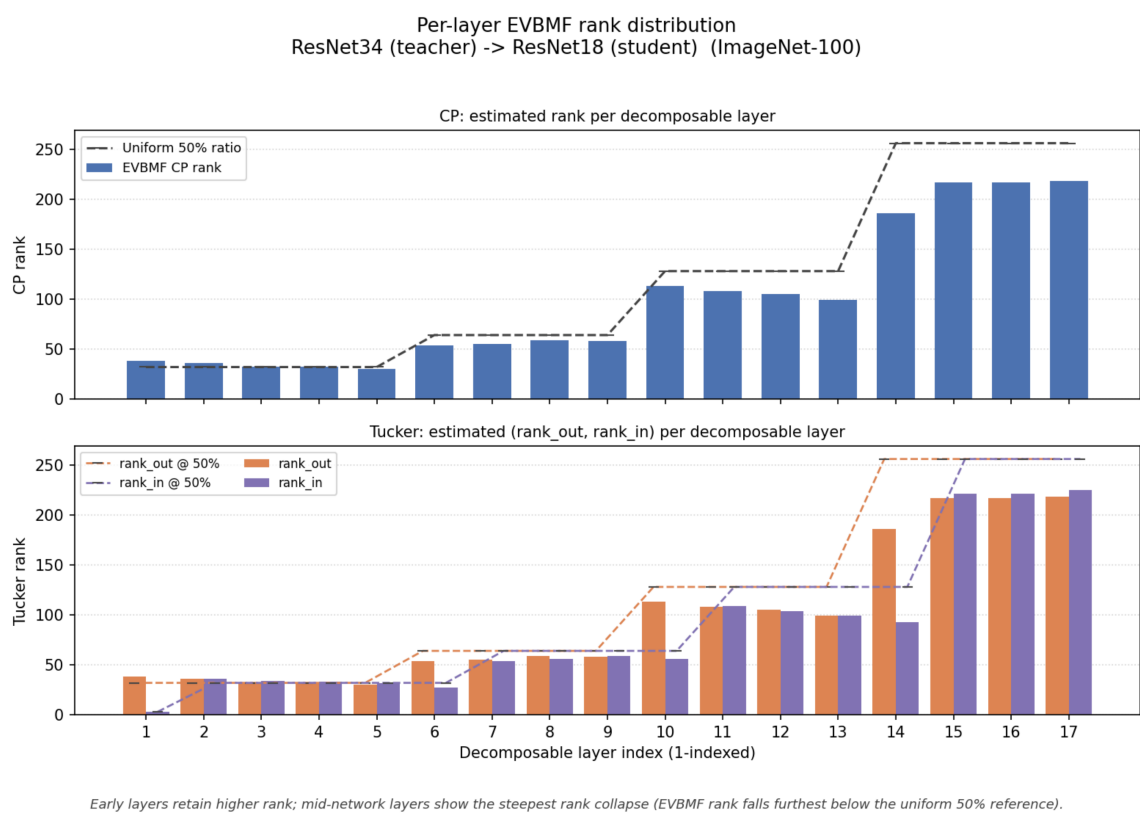
3 Contributions

- **Joint, from-scratch** construction & co-training of CP- and Tucker-decomposed students under one shared teacher — replacing the fragile decompose-then-fine-tune pipeline.
- **PURSUhInT** clustering integrated into compressed-student training to pick non-redundant, semantic hint points.
- **EVBMF** teacher-guided per-layer ranks — automatic, calibrated to the teacher's own weight spectrum.
- **Systematic study**: 8 teacher–student pairs, 2 datasets, 5 distillation objectives, one identical pipeline.

4 Automatic per-layer ranks (EVBMF)

EVBMF reads the teacher's **trained weight spectrum** to set each layer's rank — no global ratio, no validation search.

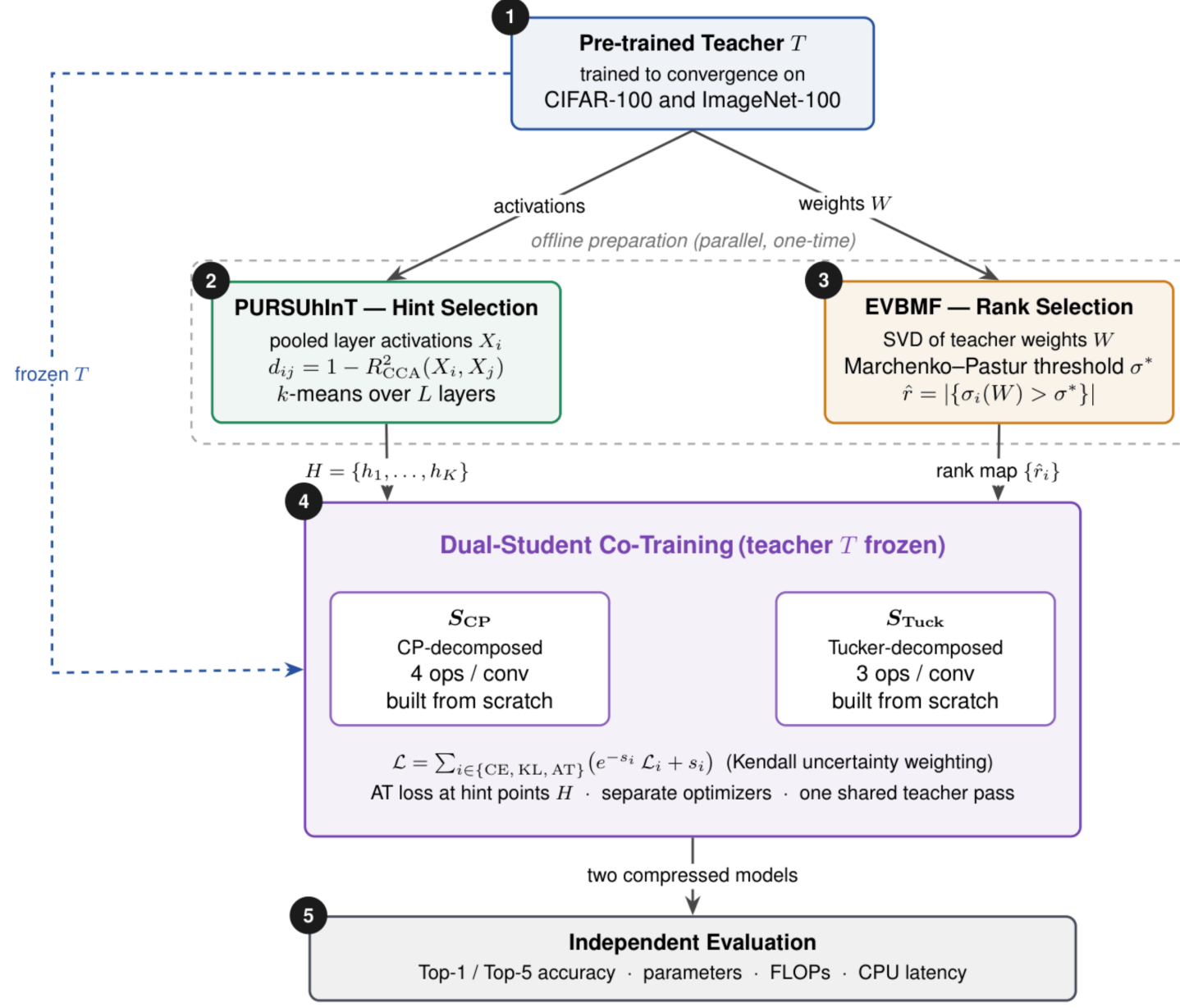
$$\hat{R} = \{ \sigma_i(\mathcal{W}) : \sigma_i(\mathcal{W}) > \sigma^*(\mathcal{W}) \}$$



Markedly non-uniform: early layers keep ~59% of channels, deep layers fall to 36–44%. Tucker estimates input/output ranks *independently* per stage (Marchenko–Pastur threshold).

5 Proposed Framework

A single joint optimization replaces the sequential pipeline: two compressed students are **built from scratch** and **co-trained** under one **frozen** teacher, then evaluated independently.



One shared teacher forward pass supervises both students each step — roughly **twice** the cost of a single distillation, not three times.

6 How it works

PURSUhInT | where to supervise

Cluster teacher layers by representational similarity; each cluster centroid is a non-redundant hint point ($K=3$ CIFAR, $K=4$ ImageNet).

$$d(i, j) = 1 - R^2_{CCA}(X_i, X_j) \xrightarrow{k\text{-means}} \mathcal{H} = \{h_1, \dots, h_K\}$$

Heterogeneous students | what to build

Same base net, factorized two ways from the EVBMF rank map:

CP (4 ops): $1 \times 1 \rightarrow \text{dw } k_H \rightarrow \text{dw } k_W \rightarrow 1 \times 1$; params $\rightarrow R(C_{in} + k_H + k_W + C_{out})$. Highest compression.

Tucker (3 ops): $1 \times 1 \rightarrow k_H \times k_W \text{ core} \rightarrow 1 \times 1$; dense core keeps cross-channel correlations $\Rightarrow$ higher accuracy *and* lower latency.

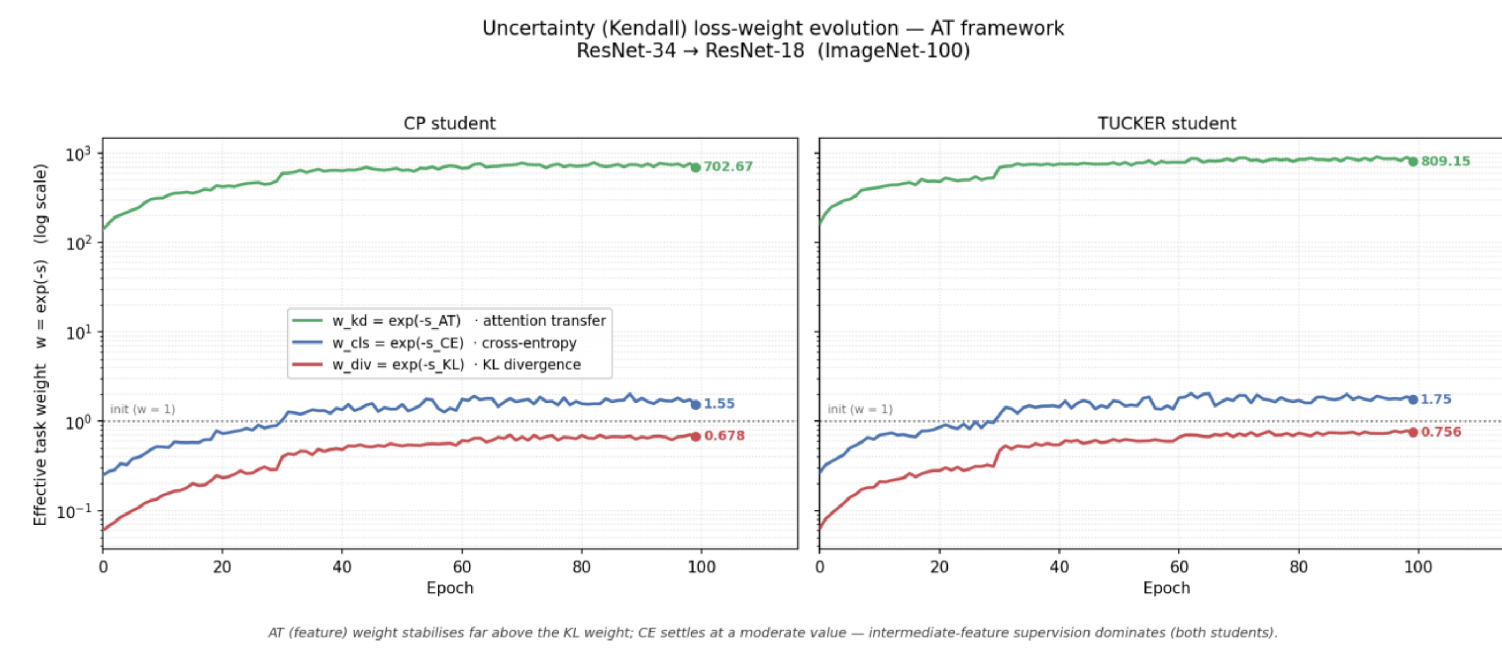
Distillation + auto-balancing | how to train

KD + attention transfer at the hint points:

$$\mathcal{L}_{KL} = \text{KL}(\sigma(z_t/\tau) \parallel \sigma(z_s/\tau)), \quad \mathcal{L}_{AT} = \sum_k \text{MSE}(a_s^k, a_t^k)$$

Homoscedastic (Kendall) uncertainty weighting — **no** hand-tuned coefficients:

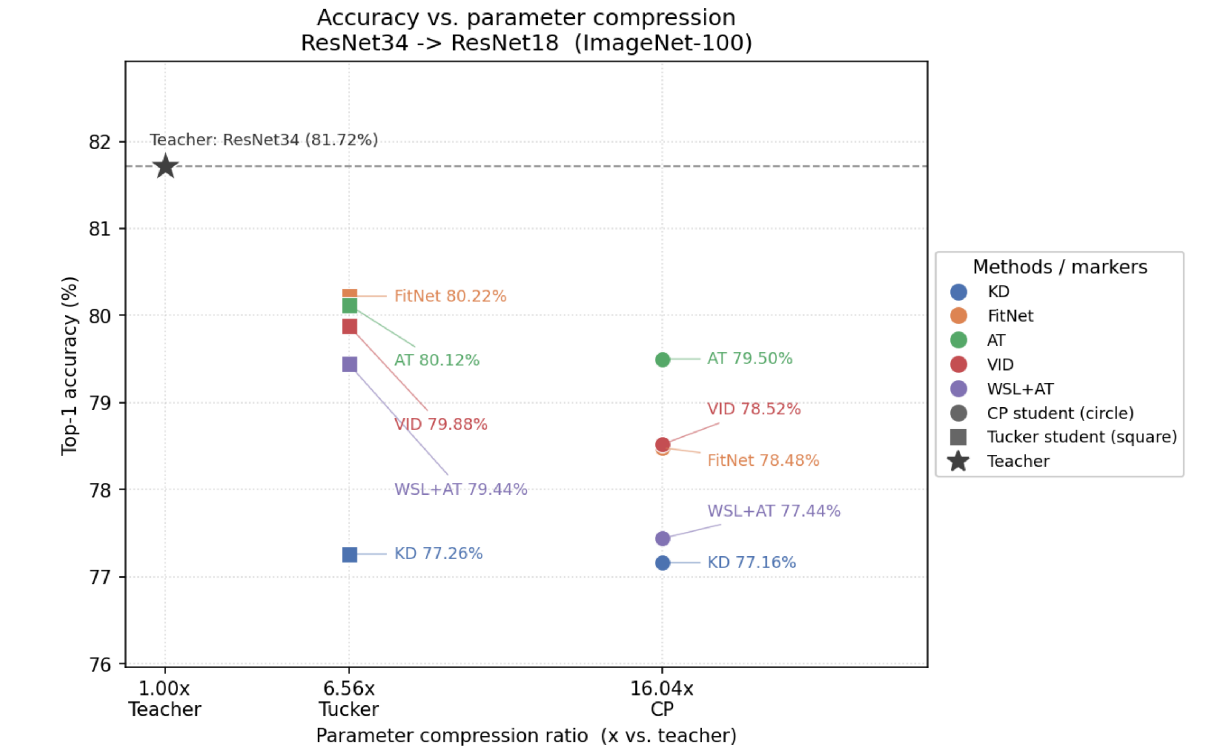
$$\mathcal{L} = \sum_{i \in \{CE, KL, AT\}} (e^{-s_i} \mathcal{L}_i + s_i)$$



Weights span **3 orders of magnitude** yet equalize the three gradient contributions to within 10% (CE 0.90, KL 1.00, AT 0.91).

7 Key Results

8 teacher–student pairs on CIFAR-100 & ImageNet-100, each under **5 distillation objectives** in one identical pipeline.



Teacher → Student	Dec.	Compr.	Top-1
<i>ImageNet-100</i>			
ResNet-34 → 18	Tucker	6.56×	80.12
	CP	16.04×	79.50
ResNet-50 → 18	Tucker	7.46×	81.04
	CP	17.59×	79.60
<i>CIFAR-100</i>			
ResNet-32×4 → WRN-16-2	CP	64.04×	72.01
WRN-40-2 → WRN-16-2	Tucker	6.78×	72.23

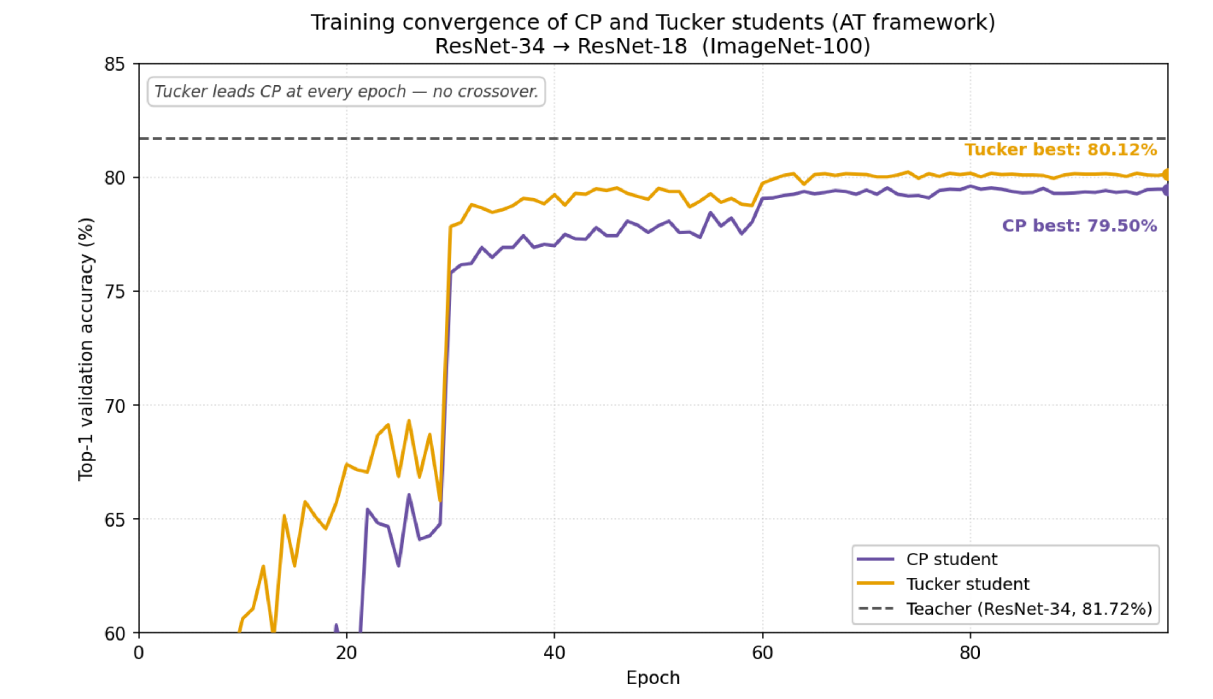
Attention transfer is best in **10 of 16** columns; feature-based supervision beats logit-only KD in *every* ImageNet config.

8 Efficiency & CP–Tucker trade-off

The two students form a **consistent trade-off** — the ordering never reverses.

	CP	Tucker
Compression	up to 64×	6–22×
Accuracy	lower	higher
CPU latency	1.96×	2.11×

Single-image CPU latency 265.98 $\rightarrow$ 135.88 ms (CP) $\rightarrow$ **125.91 ms** (Tucker). Tucker is faster *despite* 2.4× more params — a 3-op vs 4-op kernel chain.



9 Conclusions

- Joint compression + distillation **beats sequential pipelines** at every ratio tested — Tucker within **1.6 pp** of the teacher on ImageNet-100; both students within **3–5 pp** on moderate-compression CIFAR-100 pairs.
- EVBMF + PURSUhInT + Kendall $\Rightarrow$ **zero tuned hyperparameters**.
- **Tucker** for accuracy & lowest latency; **CP** for the tightest budget.
- Architecture-agnostic; complies with ISO/IEC 23053:2022 & TS 4213:2022.

Future: vision transformers · full ImageNet-1k · post-training quantization · on-device energy.

References

Please see the full reference list, notation details, additional experiments, and further explanation via the QR code.

Source code & report:

github.com/efkirmizi/distill

